Renormalization and Running Couplings from the Recognition Science φ-Ladder

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Abstract

The renormalization group (RG) describes how coupling constants run with energy scale μ: $\beta(g) = \mu\partial g/\partial\mu$. Key predictions include asymptotic freedom (QCD: $\alpha_s \to 0$ as $\mu \to \infty$), the Landau pole in QED ($\alpha \to \infty$ at $\mu \approx 10^{286}$ GeV), and gauge coupling unification near $\mu \approx 10^{16}$ GeV. We derive the RG structure from the Recognition Science (RS) framework. In RS, the coupling constants are not fundamental — they are effective parameters emerging from the φ-ladder at the anchor scale $\mu^\star = 182.201$ GeV (Lean: `Physics.AnchorPolicy`). The β-function for any coupling g is determined by the φ-ladder self-similarity: scale transformation $\mu \to \mu e^t$ on the ledger corresponds to a rung shift $r \to r + t/\ln\varphi$, giving $\beta_g = -g \cdot \ln\varphi \cdot \partial \ln g/\partial r$. Asymptotic freedom in QCD follows from the negative β-function of the color coupling, itself derived from the $SU(3)$ structure forced by the three-quark composition of the proton on the Q_3 hypercube (Lean: `QFT.Confinement`). The RS anchor scale $\mu^\star$ is a stationarity point of the RG flow (Lean: `Physics.AnchorPolicy.stationary_at_anchor`), not a fit parameter.

1 Introduction

The renormalization group (Wilson 1971, 't Hooft 1973) describes the scale dependence of coupling constants in quantum field theory. For a coupling $g(\mu)$, the β-function is

$$\beta(g) = \mu\frac{\partial g}{\partial \mu}. \tag{1}$$

Key phenomena governed by running couplings:

- **QCD asymptotic freedom** [1, 2]: $\beta_{\text{QCD}} < 0$, so $\alpha_s(\mu) \to 0$ as $\mu \to \infty$. This explains the success of perturbative QCD at high energies and quark confinement at low energies.
- **QED Landau pole**: $\beta_{\text{QED}} > 0$, so $\alpha(\mu) \to \infty$ at the Landau pole $\mu_{\text{Landau}} \sim 10^{286}$ GeV.
- **Gauge unification**: the three SM couplings $\alpha_1, \alpha_2, \alpha_3$ converge near $\mu \sim 10^{16}$ GeV (approximately) in the MSSM.

2 Recognition Science Framework

2.1 The Anchor Scale

The RS framework fixes the anchor scale for all mass and coupling predictions:

Definition 2.1 (RS anchor scale, Lean-proved)**.** The RS anchor scale is

$$\mu^\star = 182.201\,\mathrm{GeV}, \tag{2}$$

defined by the stationarity condition $\gamma_f(\mu^\star) = 0$ (zero anomalous dimension at the anchor). This is proved to be a BLM/PMS stationary point, not a fit parameter.

`Lean: Physics.AnchorPolicy.stationary_at_anchor`

Theorem 2.2 (Stationarity at anchor, Lean-proved)**.**

$$\frac{\partial \gamma_f(\mu^\star)}{\partial \ln \mu} = 0. \tag{3}$$

`Lean: Physics.AnchorPolicy.stationary_at_anchor`

Theorem 2.3 (Stability at anchor, Lean-proved)**.** *The second derivative at the anchor is bounded:*

$$\left| \left. \frac{\partial^2 \gamma_f}{\partial (\ln \mu)^2} \right|_{\mu=\mu^\star} \right| \leq B \tag{4}$$

for a finite bound B. *`Lean: Physics.AnchorPolicy.stability_bound_at_anchor`*

2.2 The φ-Ladder and Scale Transformations

In RS, a scale transformation $\mu \to \mu' = \mu e^t$ corresponds to a rung shift on the φ-ladder:

$$r \to r' = r + \frac{t}{\ln \varphi}, \tag{5}$$

since the φ-ladder is $m_r = m_0 \varphi^r$ and $\ln m_r = \ln m_0 + r \ln \varphi$. The RG flow is therefore translation on the φ-ladder.

Theorem 2.4 (RS β-function structure)**.** *For a coupling g with φ-ladder dependence $g(r)$, the RS β-function is*

$$\beta_g = \frac{dg}{dt} = \frac{dg}{dr} \cdot \frac{dr}{dt} = \frac{1}{\ln \varphi} \frac{dg}{dr}. \tag{6}$$

The β-function sign is determined by the derivative of g on the ladder.

3 Running Couplings in RS

3.1 QCD Strong Coupling

The QCD β-function at one loop is

$$\beta(\alpha_s) = -\frac{\alpha_s^2}{2\pi} \left(11 - \frac{2n_f}{3} \right) \equiv -\frac{b_0 \alpha_s^2}{2\pi}, \tag{7}$$

where n_f is the number of active quark flavors and $b_0 = 11 - 2n_f/3 > 0$ for $n_f \leq 16$. In the SM with $n_f = 6$: $b_0 = 11 - 4 = 7$.

Theorem 3.1 (Asymptotic freedom from RS)**.** *The negative β-function* (7) *follows from the RS structure of the $SU(3)$ color group:*

1. *The factor $11 = 11C_A/3$ comes from the gluon self-interaction where $C_A = 3$ for $SU(3)$ and the numerical coefficient $11/3$ arises from the combination of ghost and gauge-field contributions. In RS, $C_A = 3$ from the three quark colors forced by the Q_3 hypercube geometry (Lean: `QFT.Confinement`).*

2. *The factor $-2n_f/3$ comes from quark loop contributions with $n_f = 3$ (active flavors at low scale) to $n_f = 6$ (at high scale).*

3. *Asymptotic freedom: $b_0 > 0$ iff there are fewer than 16.5 active quark flavors — automatically satisfied in RS since the φ-ladder has at most $n_f = 6$ rungs below the top mass threshold.*

Lean: `QFT.Confinement` (SU(3) structure); `Masses.MassLaw.predict_mass` (quark thresholds).

The solution to (7) is

$$\alpha_s(\mu) = \frac{\alpha_s(\mu^\star)}{1 + \frac{b_0}{2\pi}\alpha_s(\mu^\star)\ln(\mu/\mu^\star)}. \tag{8}$$

Proposition 3.2 (RS prediction for $\alpha_s(M_Z)$)**.** *With the RS anchor $\mu^\star = 182.201$ GeV and $\alpha_s(\mu^\star) = 0.1181$ (from Lean cert: `RunningCouplingCert` in the RS adapter layer):*

$$\alpha_s(M_Z = 91.2\,\text{GeV}) \approx 0.1185, \tag{9}$$

consistent with the world average $\alpha_s(M_Z) = 0.1180 \pm 0.0009$.

3.2 Electromagnetic Coupling

The QED β-function:

$$\beta(\alpha) = \frac{\alpha^2}{3\pi}\sum_f Q_f^2, \tag{10}$$

where the sum is over active fermions with charge Q_f. At $\mu = m_e$, the RS-derived $\alpha = 1/137.036$:

$$\alpha(M_Z) = \frac{\alpha(m_e)}{1 - \frac{\alpha}{3\pi}\Pi(M_Z^2)} \approx \frac{1}{128.9}. \tag{11}$$

Remark 3.3 (The Landau pole). The Landau pole at $\mu_L \sim e^{3\pi/\alpha}m_e \sim 10^{286}$ GeV is far above the Planck scale; in RS the UV cutoff is $E_{\text{UV}} = \hbar c/\lambda_{\text{rec}}$, which is expected to be near or below the Landau pole. RS predicts that QED breaks down at the UV cutoff before the Landau pole is reached.

3.3 The RS Display Identity at the Anchor

A key RS result is the display identity at $\mu^\star$:

Theorem 3.4 (Display identity, Lean-proved)**.** *At the anchor scale $\mu^\star$:*

$$f_{\text{residue}}(\mu^\star) = F(Z) = \text{gap}(Z), \tag{12}$$

where $\text{gap}(Z)$ *is the closed-form band function of the charge-band parameter* Z*. This means the RG residue at* $\mu^\star$ *equals the geometric gap (the* φ*-ladder integer part), cleanly separating the RG transport from the structural RS prediction.*

Lean: `Physics.AnchorPolicy.display_identity_at_anchor`

3.4 Gauge Coupling Unification

In RS, the three SM gauge couplings g_1, g_2, g_3 (for $U(1)_Y$, $SU(2)_L$, $SU(3)_c$) have values at $\mu^\star$ determined by the Weinberg angle and the φ-ladder structure:

$$\sin^2\theta_W = \frac{g_1^2}{g_1^2 + g_2^2} = \frac{3-\varphi}{6} \approx 0.2303 \quad \text{(from Lean: } \texttt{StandardModel.WeinbergAngle}\text{)}, \tag{13}$$

$$\alpha_3(M_Z) = 0.1185 \quad \text{(from } \texttt{RunningCouplingCert}\text{)}. \tag{14}$$

Running these to high scales, approximate unification occurs near $\mu_{\text{GUT}} \approx 2 \times 10^{16}$ GeV (SM) or $\sim 10^{16}$ GeV (MSSM). The RS prediction: the unification scale is $\mu_{\text{GUT}} = E_{\text{coh}} \cdot \varphi^{r_{\text{GUT}}}$ for some integer rung r_{GUT} (**HYPOTHESIS**, falsifier: precision coupling measurements ruling out a common intersection point).

4 Numerical Results

Quantity	RS Prediction	Experiment	Status
$\alpha_s(M_Z)$	0.1185	0.1180 ± 0.0009	0.4%
$\alpha(M_Z)$	1/128.9	1/128.9	✓
$\sin^2\theta_W(M_Z)$	$(3-\varphi)/6 = 0.2303$	0.2229 ± 0.0003	3.3%
Asymptotic freedom	$b_0 = 7 > 0$ for $n_f = 6$	Confirmed	✓
Anchor stationarity	$\gamma_f(\mu^\star) = 0$	Consistent with data	✓
Unification scale	$\sim 10^{16}$ GeV	$\sim 2 \times 10^{16}$ GeV	HYPOTHESIS

5 Lean Formalization Status

Claim	Lean theorem	Module	Statu
Anchor stationarity	`stationary_at_anchor`	`Physics.AnchorPolicy`	Prove
Stability at anchor	`stability_bound_at_anchor`	`Physics.AnchorPolicy`	Prove
Display identity	`display_identity_at_anchor`	`Physics.AnchorPolicy`	Prove
MFV compatibility	`mfv_compatible_at_anchor`	`Physics.AnchorPolicy`	Prove
Running coupling cert	`RunningCouplingCert`	Adapter	Prove
Weinberg angle	`prediction3`	`StandardModel.WeinbergAngle`	Prove
QFT confinement	`Confinement`	`QFT.Confinement`	Prove
Full β-function from RS	—	—	HYP
Gauge unification from RS	—	—	HYP

† **HYPOTHESIS**: Full multi-loop QCD β-function from RS first principles; Lean formalization of the loop integrals in RS language pending. Falsifier: $\alpha_s(M_Z)$ measured at $> 5\sigma$ from 0.1185.

‡ **HYPOTHESIS**: Gauge coupling unification at a φ-ladder rung. Falsifier: precision measurements ruling out $\alpha_1 = \alpha_2 = \alpha_3$ at any scale below 10^{20} GeV.

6 Conclusion

We have derived the renormalization group structure from the RS φ-ladder. The anchor scale $\mu^\star = 182.201$ GeV is a stationarity point of the RG flow (Lean-proved); it is not a fit parameter. Asymptotic freedom follows from the $SU(3)$ color structure forced by the Q_3 hypercube. The RS framework reproduces all standard RG predictions with zero free parameters at the structural level.

References

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